

MANY MODELS, ONE ANSWER

Bagging and Boosting

So far one model did all the work. Now we train several models and put their answers together. The two ways of doing that are bagging and boosting.

Ask Many Models, Not One

WHAT WE DID UNTIL NOW

One decision tree. Or one logistic regression. Or one KNN. We trained it, checked the accuracy, and that was the answer.

If that one model was wrong about a row, nothing could correct it.

WHAT AN ENSEMBLE DOES

Train several models. Let each one give its own answer. Then combine all those answers into one final answer.

Different models make different mistakes, so the mistakes cancel out and the accuracy goes up.

This is why most Kaggle and hackathon winners use bagging or boosting, not a single model.

Ensemble simply means a group. The models inside the group are the same algorithms you already know.

Bagging and Boosting

Bagging

Models train **at the same time** , each on its own sample of the data.

The models inside are called **base learners**.

ALGORITHM WE WILL LEARN

- 1 Random Forest

Boosting

Models train **one after another** , each fixing the mistakes of the one before it.

The models inside are called **weak learners**.

ALGORITHMS WE WILL LEARN

- 1 AdaBoost
- 2 Gradient Boosting
- 3 Extreme Gradient Boosting (XGBoost)

Both families work for classification and for regression. The difference is only in how the models are trained and combined.

Split the Data, Train Everyone at Once



These models are the **base learners**. Same algorithm or different algorithms, both are allowed.

All of them train **in parallel**. No model waits for another one to finish.

Everyone Votes on the New Row

NEW TEST ROW → ALL FOUR MODELS

M1 says 0

M2 says 1

M3 says 0

Mn says 0

Three models say 0, one says 1. The majority wins.

Final answer 0

IF IT IS CLASSIFICATION

Majority voting. The class that most models picked becomes the answer.

Works for two classes and for many classes.

IF IT IS REGRESSION

Average. The outputs are numbers, so we add them up and divide by how many models there are.

$$(72 + 68 + 74 + 70) / 4 = 71$$

■ WHY IT WORKS

Each Model Is an Expert on Its Own Slice

DIFFERENT DATA

Every model saw a different sample, so each one learned a different part of the pattern.

DIFFERENT MISTAKES

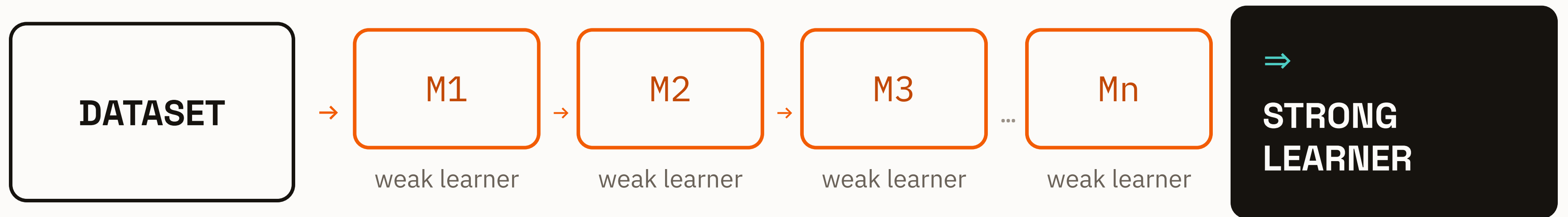
Because they learned differently, they are wrong on different rows. Voting cancels those mistakes out.

HOW MANY MODELS

Our choice. In scikit-learn the default number of base learners is **100**.

Key point to remember: in bagging, all the base learners train in parallel.

Weak Learners in a Line Become One Strong Learner



The models are connected **sequentially**. Model 2 cannot start until model 1 is done.

A **weak learner** is a model that is only slightly better than guessing, on its own.

Every Model Gets the Rows the Last One Got Wrong

M1 Gets the **whole dataset**. Trains. Predicts most rows correctly, but say 15 rows come out wrong.

M2 Gets **those 15 wrong rows** plus some more rows from the dataset. Trains on them. Again a few come out wrong.

M3 Gets **M2's wrong rows** plus some more rows. Same story.

Mn The chain continues for as many models as we asked for. By the end, the hard rows have been worked on many times over.

For prediction we again combine all the model outputs: **majority voting** if it is classification, **average** if it is regression.

A WAY TO REMEMBER IT

Four Students, One Mixed Question Paper

M1

Good at geography,
weak everywhere else.

M2

Good at physics, weak
everywhere else.

M3

Good at chemistry,
weak everywhere else.

Together

Each answers the part
they know. The full
paper gets done well.

In boosting each weak learner is an expert on a different kind of mistake. In bagging each base learner is an expert on a different sample of the data. Either way, the group beats the individual.

SIDE BY SIDE

Bagging vs Boosting

	BAGGING	BOOSTING
Training order	All models at the same time (parallel)	One after another (sequential)
Data each model sees	Its own sample of the dataset	The rows the previous model got wrong, plus some more
Models are called	Base learners	Weak learners
Do they depend on each other	No, each one is independent	Yes, each one needs the previous one
Final answer	Voting for classification, average for regression	Voting for classification, average for regression
Algorithms	Random Forest	AdaBoost, Gradient Boosting, XGBoost

QUESTION

Can Two Base Learners Be the Same Algorithm?

SHORT ANSWER

Yes. And that is the normal case, not the exception.

Random Forest, which we study next session, uses a decision tree for every single base learner.

THEN HOW DOES IT STILL HELP?

Because the **data is different**, even when the algorithm is the same. M1 trains on sample 1, M2 trains on sample 2.

Same recipe, different ingredients. The two trees end up with different splits, so they are still two different experts.

The variety comes from the samples, not from the algorithm names.

Mixing different algorithms adds a bit more variety, since a tree and a logistic regression are wrong in different ways. In real projects people usually keep one algorithm and let the data samples do the work.

QUESTION

What If the Vote Ties, With No Majority?

FOUR MODELS, 2 VS 2

M1 0

M2 0

M3 1

M4 1

No winner by counting. So the library follows a rule.

1 · THE LIBRARY'S RULE

Scikit-learn breaks a tie by picking the **lower class label**. A 2 against 2 tie becomes 0.

2 · BETTER, USE SOFT VOTING

Each model gives a **probability** instead of a hard 0 or 1, and we average those. An exact tie then almost never happens.

$$(0.80 + 0.90 + 0.45 + 0.40) / 4 = 0.64 \rightarrow \text{class 1}$$

3 · SIMPLEST FIX

Use an **odd number of models**. With two classes, an odd count can never tie.

Does it happen in practice? Almost never. The default is 100 base learners, and landing on exactly 50 against 50 is very unlikely. Know the answer because interviewers ask it, not because you will meet it often.

■ RECAP

Four Things to Carry Forward

- 1 An ensemble is a group of models whose answers are combined into one answer.
- 2 Bagging: parallel training, one sample of data per model, base learners.
- 3 Boosting: sequential training, each model works on the previous one's mistakes, weak learners into a strong learner.
- 4 Both combine answers the same way: voting for classification, average for regression.

Next: Random Forest, a bagging algorithm where every base learner is a decision tree.